

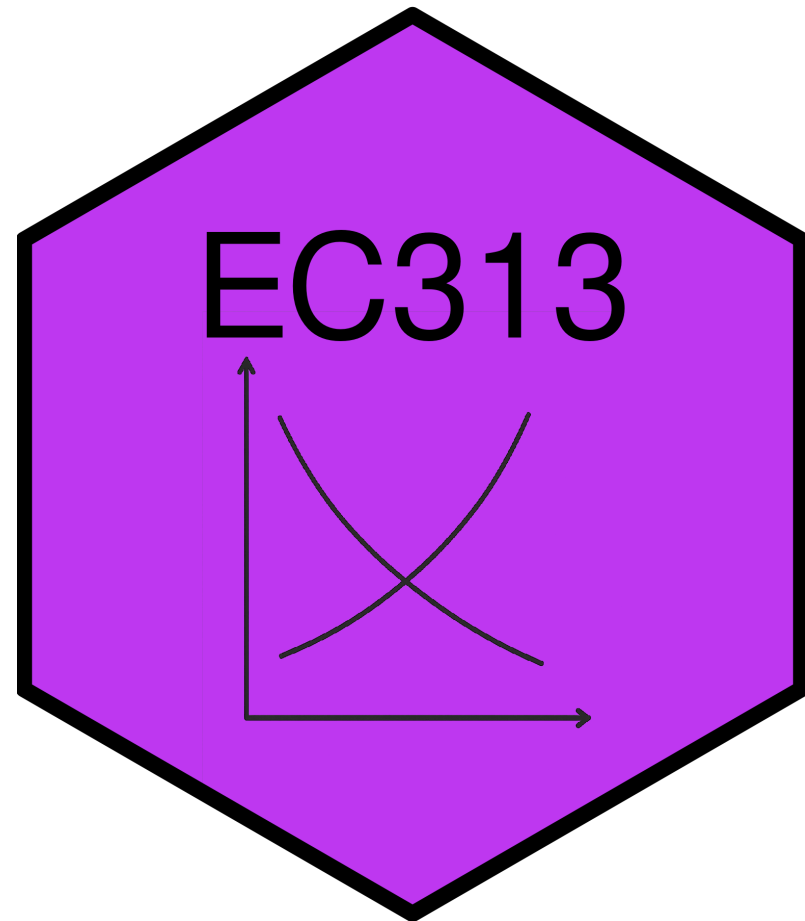
The Corporation Tax

EC313 - Public Economics: Taxation

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Fall 2026



Goals of This Section

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In this section we will:

- Provide an overview of how corporate taxes work in Canada
- Discuss the incidence and excess burden of corporate taxes
- Understand how corporate taxes affect investment and financing decisions
- Discuss the taxation of multinational corporations
- Explore ideas for corporate tax reform

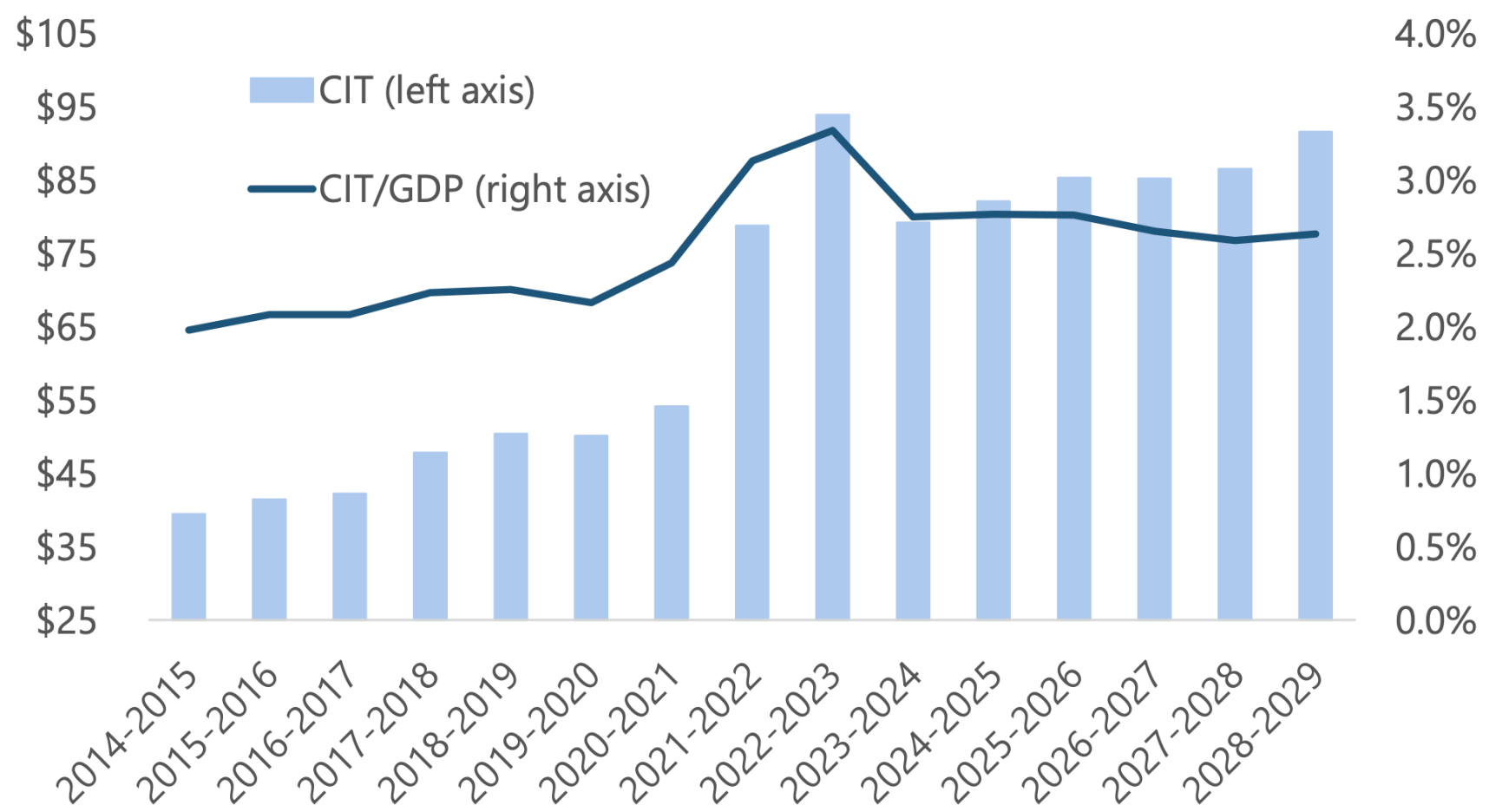
Introduction

Introduction

- So far we have discussed individual taxation
- **Corporations** also pay taxes, and are taxed separately
- A corporation is a type of business organization that
 - Is a legal entity that is separate from its owners
 - Is owned by shareholders
 - Can enter into contracts, own assets, and incur liabilities
- The structure limits **liability** for shareholders
 - Shareholders are not personally liable for the corporation's debts or legal obligations beyond their investment in the company

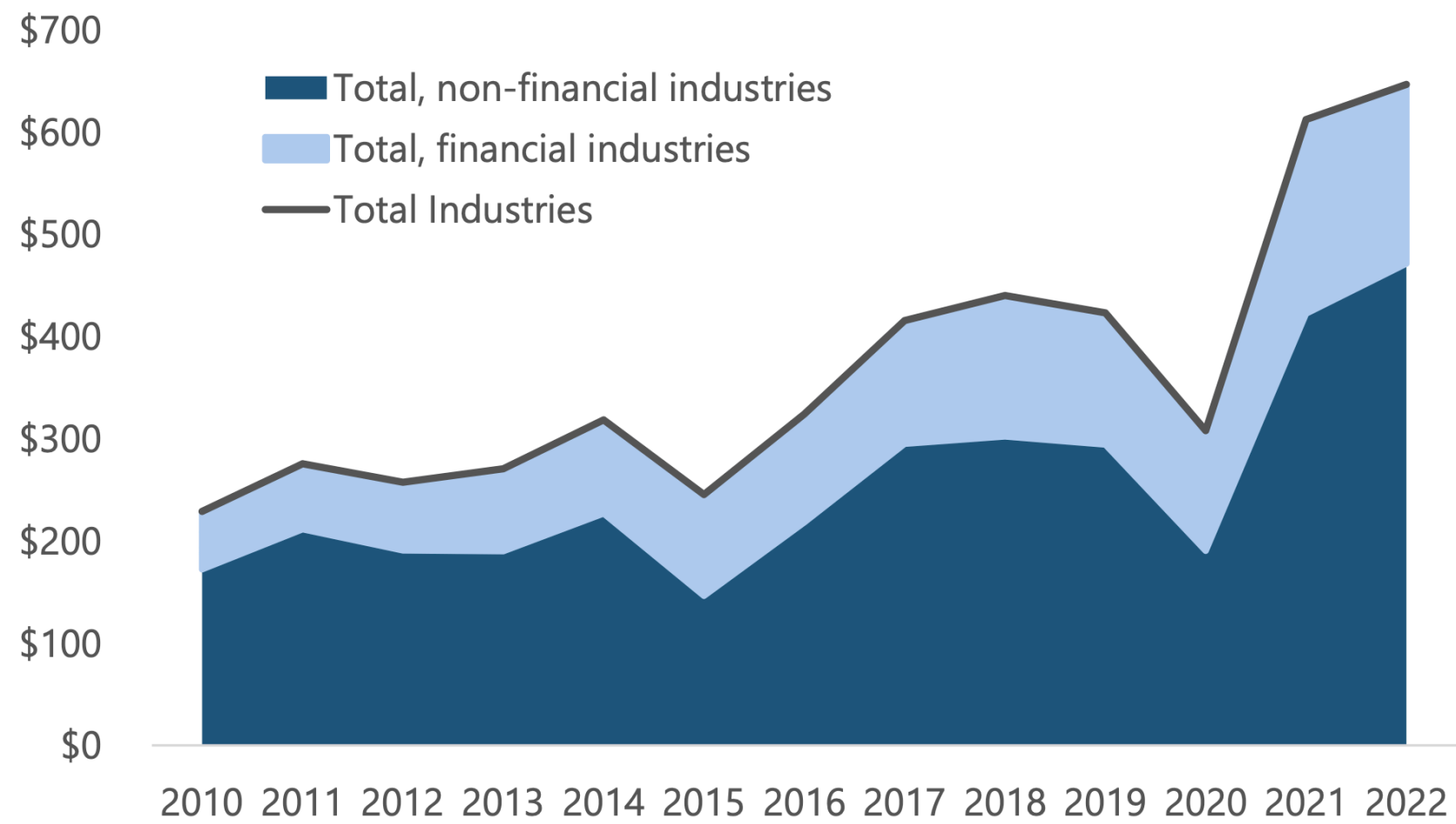
Introduction

Evolution of corporate income tax revenue from 2014-15 to 2028-29, billions of dollars and as a percentage of GDP



Introduction

Corporate income before taxes from 2010 to 2022, billions of dollars



Introduction

Income before taxes for the top five non-financial industries from 2018 to 2022

	Income 2018 (\$ billions)	Income 2022 (\$ billions)	Growth from 2018 to 2022	Share of income 2018 to 2022 (average)	Contribution to growth 2018 to 2022
Manufacturing	\$71.4	\$81.5	14.1%	13.8%	4.9%
Real estate and rental and leasing	\$39.0	\$59.1	51.4%	9.2%	9.7%
Wholesale trade	\$34.0	\$53.5	57.3%	8.3%	9.4%
Mining, quarrying, and oil and gas extraction	\$1.8	\$70.8	3,916.2%	1.3%	33.4%
Construction	\$23.5	\$38.8	65.5%	6.4%	7.4%

Introduction

Analysis of federal CIT paid for the five top paying non-financial industries from 2018 to 2022

	Federal tax paid 2018 (\$ billions)	Federal tax paid 2022 (\$ billions)	Growth from 2018 to 2022	Share of total federal tax 2018 to 2022	Contribution to growth 2018 to 2022
Manufacturing	\$6.6	\$12.8	93.5%	12.1%	22.4%
Real estate and rental and leasing	\$5.8	\$8.7	51.4%	10.1%	10.7%
Wholesale trade	\$5.4	\$7.6	40.1%	8.9%	7.9%
Professional, scientific and technical services	\$3.8	\$5.7	48.6%	7.1%	6.7%
Construction	\$3.8	\$5.6	46.0%	7.0%	6.4%

Introduction

Total share of federal tax paid by industry

	2018	2019	2020	2021	2022
Finance and insurance	27.8%	28.6%	28.8%	27.2%	22.2%
Manufacturing	11.8%	10.7%	10.4%	12.2%	15.3%
Real estate and rental and leasing	10.3%	10.3%	9.8%	9.8%	10.5%
Wholesale trade	9.7%	8.3%	8.6%	8.7%	9.1%
Professional, scientific and technical services	6.8%	6.9%	7.5%	7.5%	6.8%
Construction	6.9%	6.9%	7.5%	6.9%	6.7%
Retail trade	5.5%	5.3%	6.0%	6.3%	6.1%
Educational, health care and social assistance services	5.2%	5.5%	5.1%	4.6%	4.4%
Transportation and warehousing	3.8%	4.6%	4.1%	3.7%	4.1%
Information and cultural industries	1.3%	1.7%	1.4%	3.8%	5.3%
Agriculture, forestry, fishing and hunting	3.2%	3.5%	3.5%	2.9%	2.6%

Introduction

TABLE 2I.1

Corporate Income Taxes in Canada, 1945–2020

Year	Federal Revenue (millions)	Provincial Revenue (millions)	Federal + Provincial (millions)	As a Percent of Government Revenue (%)	Real per Capita (\$2020)
1945	\$ 645	\$ –	\$ 645	17.8	\$ 796
1955	1,081	54	1,135	16.5	703
1965	1,759	523	2,282	13.7	948
1975	5,748	2,091	7,921	12.6	1,620
1985	9,210	4,033	13,243	7.3	1,116
1995	12,432	7,093	19,525	6.1	1,043
2005	30,528	16,400	46,928	8.7	1,866
2015	41,730	26,947	68,667	9.7	2,083
2020	55,226	35,308	90,534	10.7	2,383

Sources: Adapted from the Statistics Canada publications “Historical Statistics of Canada,” 1983, Catalogue 11-516, released July 29, 1999; “Public Finance Historical Statistical Data,” 1965/66–1991/92, Catalogue 68-512, released April 1, 1992, and “Public Sector Statistics,” 2007/08, Catalogue 68-213-X, released July 14, 2008 (Tables 2-3, 2-4, 2-6); and from Statistics Canada, data table 36-10-0450-01; and Finances of the Nation, Macroeconomic Database, <https://financesofthenation.ca/macrodata/>.

Introduction

- Corporations are important in terms of economic activity
- Also in terms of the taxes they pay
- Given they pay tax, there are **efficiency** and **equity** considerations
 - We will explore these in this chapter
- Corporate taxation is a complex and politically sensitive topic
 - We will only cover the basics here

Why Tax Corporations?

Why Tax Corporations?

- In Canada, we have a personal income tax
 - The income of a corporation will eventually be distributed to individuals
 - So it could be taxed with the personal income tax
- Why have a separate tax for corporations?

Reasons for a Separate Corporate Tax

- Corporations receive **special privileges**: limited liability, perpetual life, easy transfer of ownership
- These things are valuable, and so it is reasonable to tax corporations for them

Why Tax Corporations?

- It also withholds taxes from **foreign companies** that operate in Canada
 - Without a corporate tax, foreign companies could avoid paying Canadian taxes on their Canadian income
 - The corporate tax ensures that foreign companies contribute to the cost of public goods and services in Canada
- It allows the government to capture **economic rents**
 - Some large corporations earn economic rents due to their market power or unique resources
 - Taxing these rents is a way to raise revenue from them
- It creates **fairness** with personal taxes
 - Without a corporate tax, individuals could avoid paying personal income taxes by channeling their income through corporations
 - The corporate tax helps to prevent this type of **tax avoidance**



Think About It

If corporate income is eventually distributed to individuals and taxed under the personal income tax, is the corporate tax just a form of prepayment? Could we achieve the same policy goals – capturing rents, taxing foreign firms, preventing avoidance – without a separate corporate tax? Why or why not?

How Corporate Taxes Work in Canada

Tax Rates

- Corporate tax rates are less complex than personal income tax rates
- There are three main rates:
 - General corporate tax rate
 - Small business tax rate on the first \$500,000 of active business income
 - Manufacturing and processing rate
- Income is taxed at both the federal and provincial level
 - Just like personal income

Tax Rates

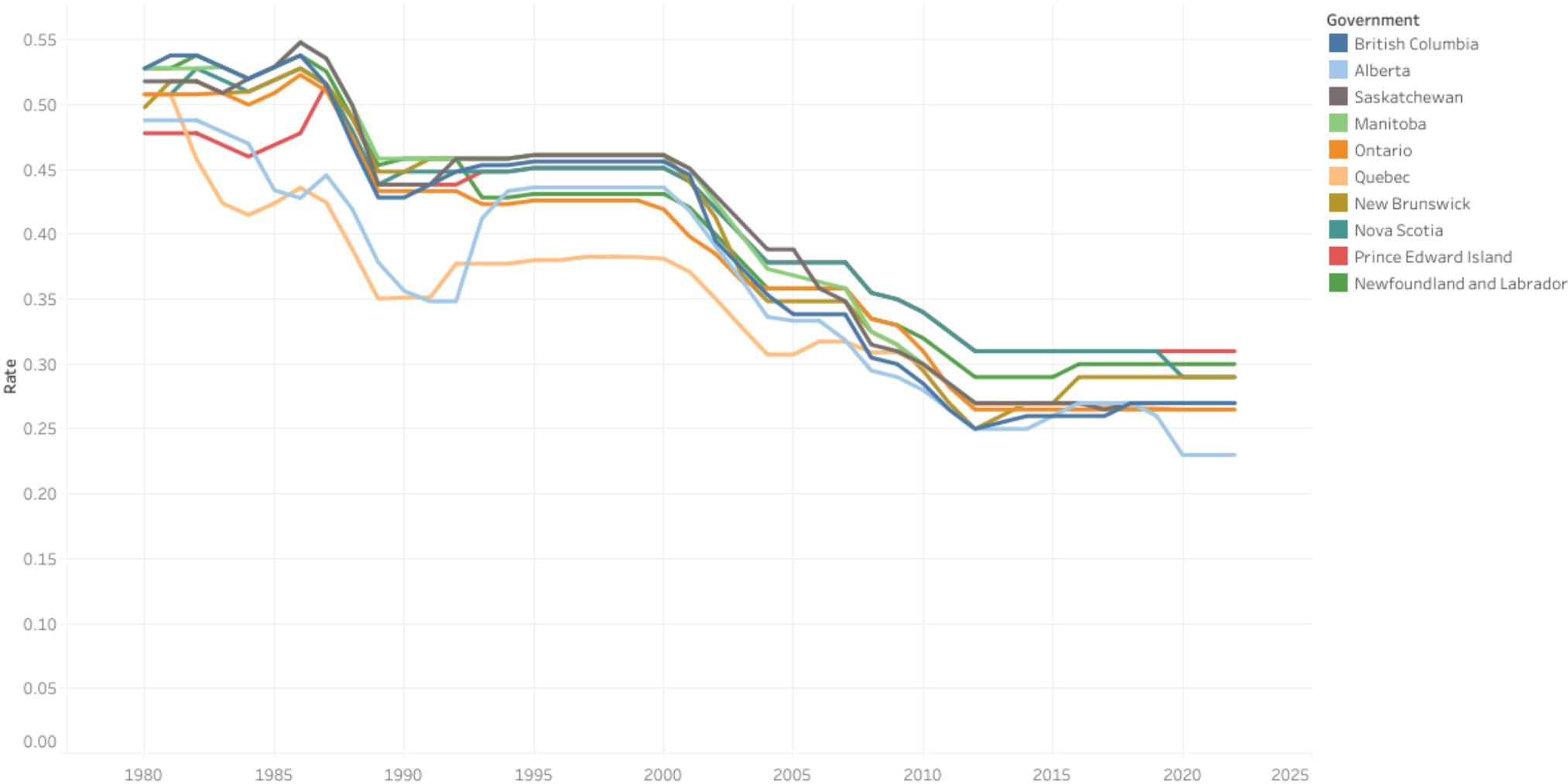
TABLE 21.2 Combined Federal and Provincial Corporate Income Tax Rates, 2022

	Small Business	Manufacturing and Processing	General
Federal	9.0%	15.0%	15.0%
Province (federal plus provincial)			
Newfoundland and Labrador	12.0	30.0	30.0
Prince Edward Island	10.0	31.0	31.0
Nova Scotia	11.5	29.0	29.0
New Brunswick	11.5	29.0	29.0
Quebec	12.2	26.5	26.5
Ontario	12.2	25.0	26.5
Manitoba	9.0	27.0	27.0
Saskatchewan	10.0	25.0	27.0
Alberta	11.0	23.0	23.0
British Columbia	11.0	27.0	27.0
Northwest Territories	13.0	26.5	26.5
Nunavut	12.0	27.0	27.0
Yukon	9.0	17.5	27.0
Unweighted average	11.1	26.4	27.4

Source: Finances of the Nation, Statutory Tax Rate Database, <https://financesofthenation.ca/statutory-tax-rates/>.

Tax Rates

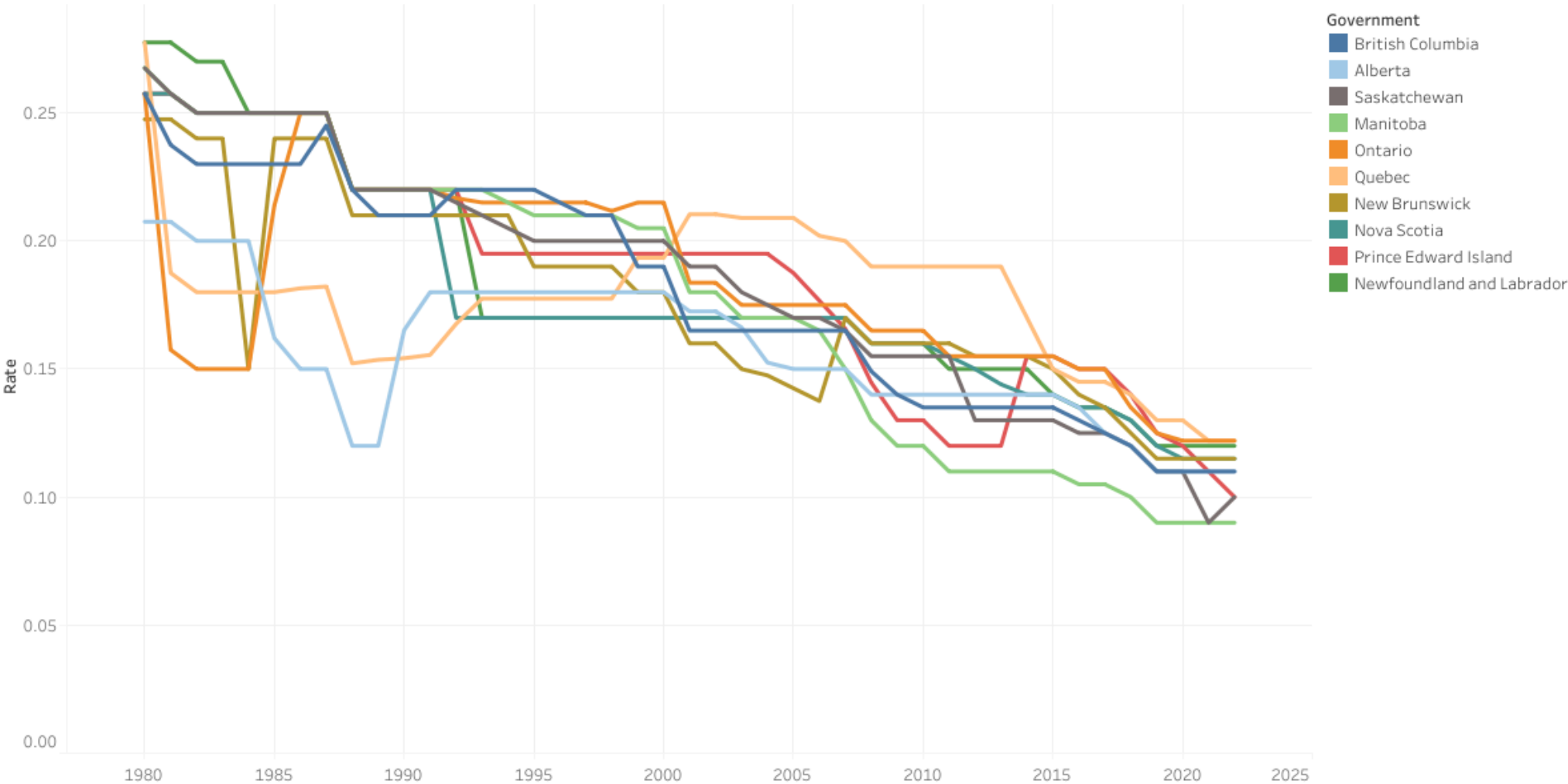
CIT General
Total tax rate



Tax Rates

CIT Low Rate

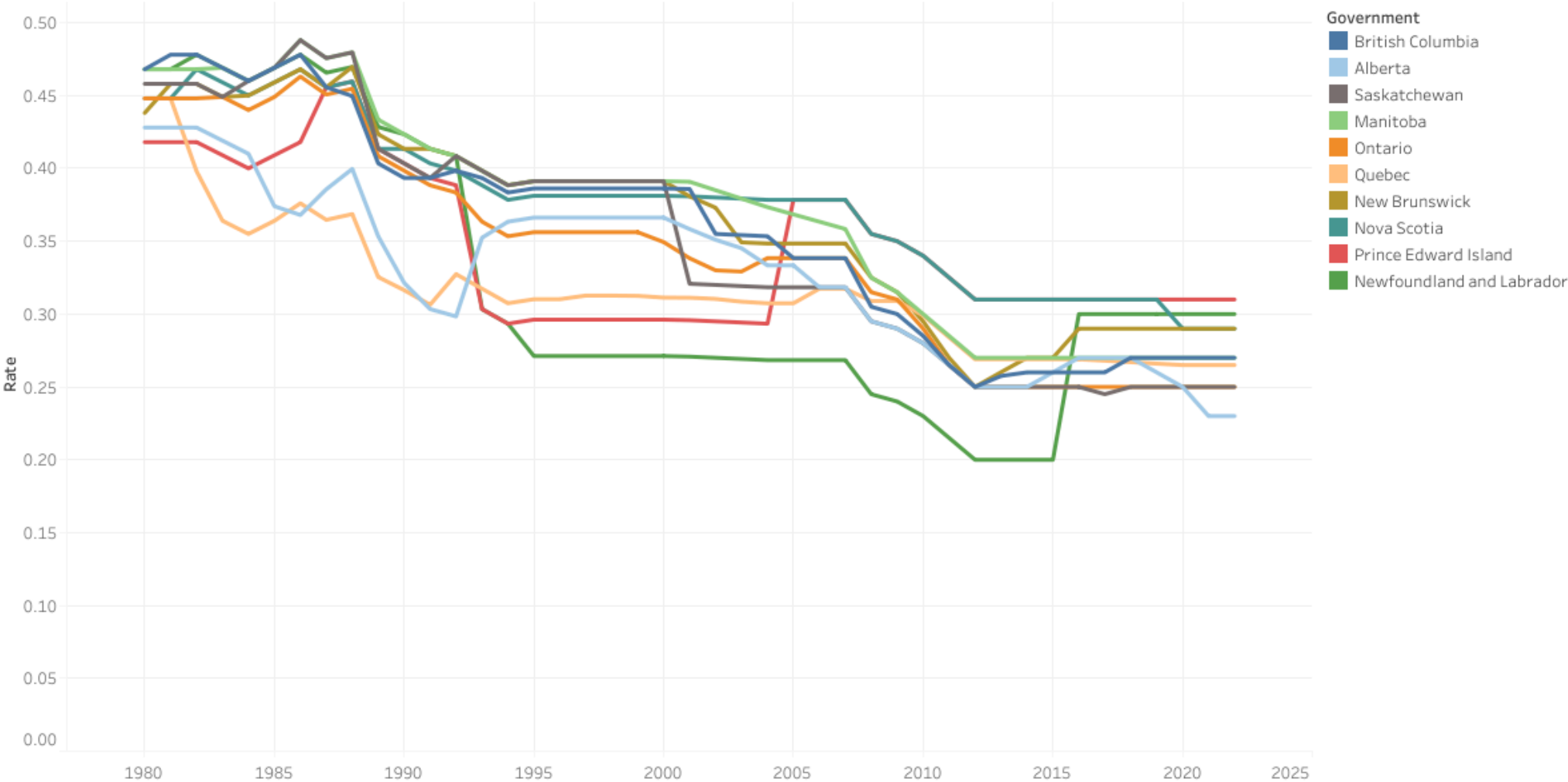
Total tax rate



Tax Rates

CIT Manufacturing & Processing

Total tax rate



Tax Rates

- These tax rates determine the **statutory incidence** of the corporate tax
- We know that statutory incidence is not the same as **economic incidence**
- Later we will explore who actually bears the burden of the corporate tax
 - As before, it is complicated
 - But generally speaking, the economic incidence falls on **capital owners** and **workers**

Wage Payments

- Corporations pay wages to workers
- This is a cost of doing business
- Wage payments to labour are **excluded** from the corporate tax base

Depreciation

- Business expenses are deductible from taxable income
- But not all expenses are the same
- Some get used up quickly (e.g., office supplies)
- Others last a long time (e.g., machinery, buildings)
- For tax purposes, they are treated differently
 - For long-lasting assets, **depreciation** (the part consumed in the year) is used to allocate the cost over time
 - The depreciation is deducted from taxable income each year

Depreciation

- How does this work in practice?
- Suppose a firm buys a machine for \$1000 that lasts 10 years
- If using the **straight line method** (deductions spread evenly), each year it can deduct \$100 from its taxable income as depreciation
- This reduces the firm's taxable income and thus its tax liability
 - If the corporate tax rate is 20%, the firm saves \$20 in taxes each year due to the depreciation deduction
- The present value of this stream of tax savings is

$$PV = \frac{20}{(1+r)} + \frac{20}{(1+r)^2} + \dots + \frac{20}{(1+r)^{10}}$$

Depreciation

- The tax deductions effectively reduce the cost of the asset to the firm

$$\text{After-tax cost} = 1000 - PV$$

- More generally, the present value of the tax savings is

$$PV = \frac{D(1) \times t \times q}{(1 + r)} + \frac{D(2) \times t \times q}{(1 + r)^2} + \dots + \frac{D(T) \times t \times q}{(1 + r)^T}$$

- where
 - $D(n)$ is the share of the initial asset cost that can be deducted in year n
 - t is the corporate tax rate
 - q is the initial cost of the asset

Depreciation

- If you factor out the initial cost q , you get

$$PV = q \times \left(\frac{D(1) \times t}{(1+r)} + \frac{D(2) \times t}{(1+r)^2} + \dots + \frac{D(T) \times t}{(1+r)^T} \right)$$

- If we call the term in brackets ψ , then the tax savings is $q \times \psi$
- The after-tax cost of the asset is then

$$\text{After-tax cost} = (1 - \psi)q$$

Depreciation

- From that equation, the tax savings are higher if
 - The total depreciation schedule is shorter (smaller T)
 - The total depreciation schedule is faster ($D(n)$ bigger in earlier years)
- This means that **accelerated depreciation** is more valuable for tax purposes
- The most valuable is **full expensing**, where the entire cost can be deducted in the year of purchase
- In Canada, certain investments are eligible for accelerated depreciation
 - Explained in detail here: <https://www.canada.ca/en/revenue-agency/services/tax/businesses/topics/sole-proprietorships-partnerships/report-business-income-expenses/claiming-capital-cost-allowance/accelerated-investment-incentive.html>

Depreciation

- An alternative to the straight line method is the [declining balance method](#)
 - Declining balance is more common in Canada
- Under declining balance, a fixed percentage of the remaining undepreciated value is deducted each year
- For example, if the declining balance rate is 20%, in the first year the firm deducts 20% of the initial cost, in the second year it deducts 20% of the remaining value after the first year's deduction, and so on
- This results in larger deductions in the earlier years and smaller deductions in later years

Investment Tax Credits

- In addition to depreciation deductions, firms may also be eligible for [investment tax credits \(ITCs\)](#)
- Recall: deductions reduce taxable income, while credits reduce tax liability directly
- ITCs are typically a percentage of the purchase price of investments
- Reduce the tax liability in the year the investment is made
- Example: a 10% ITC on a \$1000 machine would reduce the firm's tax liability by \$100 in the year of purchase
- Canada has used ITCs to encourage investment in certain sectors or regions
 - Manufacturing and processing, clean energy
 - Atlantic region

Interest

- Corporations can deduct **interest payments** on debt from their taxable income
- Suppose the firm borrowed \$1000 for 10 years at 10% interest
 - Each year, the firm pays \$100 in interest
- The firm can deduct this \$100 from its taxable income each year
 - If the corporate tax rate is 20%, saves \$20 in taxes each year

Interest

- The present value of this stream of tax savings from interest deductions is

$$PV = \frac{20}{(1+r)} + \frac{20}{(1+r)^2} + \dots + \frac{20}{(1+r)^{10}}$$

- In general, the present value of the tax savings from interest deductions is

$$PV = \frac{i \times q \times t}{(1+r)} + \frac{i \times q \times t}{(1+r)^2} + \dots + \frac{i \times q \times t}{(1+r)^T}$$

Interest

- Factoring out the initial loan value q , you get

$$PV = q \times \left(\frac{i \times t}{(1 + r)} + \frac{i \times t}{(1 + r)^2} + \dots + \frac{i \times t}{(1 + r)^T} \right)$$

- If we call the term in brackets λ , then the tax savings is $q \times \lambda$
- The after-tax cost of the loan is then

$$\text{After-tax cost} = (1 - \lambda)q$$

Dividends

- Dividends are a main way to extract profits from a corporation
 - They are payments made to shareholders from the corporation's earnings
- Dividends are taxed as personal income to the shareholders
- They are not deductible from the corporation's taxable income
- Raises the question of double taxation
 - Corporation pays tax on its earnings
 - Shareholders pay tax on dividends received

Dividends

- To mitigate double taxation, Canada has a **dividend gross-up and tax credit system**
- When shareholders receive dividends, they must “gross up” the dividend amount by a certain percentage to reflect the pre-tax income of the corporation
- They then pay personal income tax on this grossed-up amount
- To offset this, shareholders receive a **dividend tax credit** that reduces their personal tax liability
- In the end, the total tax paid on corporate earnings distributed as dividends is intended to be roughly equivalent to the tax that would have been paid if the income were earned directly by the shareholders

Dividend Tax Credit System

- Imagine a corporation earns \$125 in pre-tax profits
- It pays a corporate tax of 20%, leaving \$100 in after-tax profits
- Distributing this \$100 as a dividend to a shareholder
- The shareholder grosses up the dividend by 25% (the gross-up rate for eligible dividends), resulting in a grossed-up amount of \$125
- The shareholder pays personal income tax on this \$125
- If the rate is 40%, the tax owed is \$50

Dividend Tax Credit System

- The shareholder then receives a dividend tax credit to offset some of this tax
- If the tax credit is 20% of the grossed-up amount, the credit is \$25
- The shareholder's net tax liability on the dividend is then
 - Tax owed: \$50
 - Minus tax credit: \$25
 - Net tax liability: \$25
- Total tax paid on the original \$125 of corporate profits is
 - Corporate tax: \$25
 - Shareholder tax: \$25
 - Total tax: \$50

Dividend Tax Credit System

- Notice that this is the same as if the shareholder had earned the \$125 directly and paid 40% personal income tax
 - It instead comes partly from the corporation and partly from the shareholder
- This system helps to **integrate** corporate and personal taxation, reducing the double taxation of dividends
- Currently in Canada for regular businesses
 - The federal gross-up rate for eligible dividends is 38%
 - The federal dividend tax credit rate is 15.02% of the grossed-up amount
- For small businesses, the rates are different
 - The federal gross-up rate is 15%
 - The federal dividend tax credit rate is 9.03% of the grossed-up amount
- There are also provincial variations

Think About It

The dividend gross-up and tax credit system is designed to make the total tax on corporate earnings roughly equal to what individuals would pay if they earned the income directly. But does this integration work equally well for shareholders in different tax brackets? Who benefits most from any imperfections in the system?

Retained Earnings

- Corporations can **retain earnings** instead of distributing them as dividends
- Retained earnings are reinvested in the business or held as cash reserves
- Should a firm retain earnings or pay them out as dividends?

If given out as dividends

- Shareholders pay personal income tax on the dividends received
- Recall that because of the gross-up and tax credit system, this is roughly equivalent to personal income tax

If retained

- No personal income tax
- But value of shares increase
- Shareholders pay capital gains, but only when they sell the shares
- Inclusion rate is 50%

Retained Earnings

- Preferential treatment of capital gains creates an incentive to retain earnings

Tax Expenditures

Definition

Tax expenditures are indirect ways the government spends money through the tax system – through deductions, credits, exemptions, and preferential rates.

- Governments use both the personal and corporate tax systems for tax expenditures
- Examples in the corporate tax system include
 - Charitable donation deduction
 - Film tax credit
 - Investment tax credit
 - Lower small business tax rates
- All ways in which government targets assistance to particular activities or groups

Effective Tax Rates

- The [statutory corporate tax rate](#) is not the same as the [effective tax rate](#)
- The effective tax rate takes into account deductions, credits, inflation, and other provisions that affect the actual tax paid
- The combined federal + provincial statutory rate averages around 28%
- The average effective tax rates vary from 7%-23%
 - Higher for larger firms and retail operations
 - Lowest for firms in forestry and manufacturing

Effective Tax Rates

- **Inflation** is one factor that can change the effective tax rate
- Three main ways inflation affects corporate taxes
 - Undervalues deductions for goods in inventory, leads to **higher taxable income**
 - Depreciation is based on historical cost, not replacement cost, leading to **higher taxable income**
 - Interest deductions lead to **lower taxable income**
 - Deduction based on nominal interest, which includes an inflation component
 - Inflation component is not a real cost, so deduction is overstated

Effective Tax Rates

- Overall, inflation tends to **increase** the effective corporate tax rate
- Studies show that adjusting for inflation can reduce corporate income tax substantially
 - Old studies showed that they would be about 40% lower
 - Strongest effects in capital-intensive industries like forestry
 - For some companies tax on real profits was greater than 100%

Incidence and Excess Burden

Introduction

- Two key concepts we have come back to throughout the course are
 - **Incidence**: who ultimately bears the burden of a tax
 - **Excess burden**: the efficiency cost of a tax beyond the revenue raised
- We discussed in the context of consumption and personal income taxes
- Corporate tax can also create these issues

Tax Incidence

- Suppose the government levies a corporate tax
 - The **statutory incidence** is on the corporation
- Who bears the **economic incidence**?
- If the corporation produces goods/services, it may shift some to **consumers**
 - Higher prices
 - Depends on price elasticity of demand

Tax Incidence

- Suppose the corporations bear some of the economic burden
- In a general equilibrium setting, they can shift to other factors of production
 - Labour
 - Capital
- If corporations employ a large share of workers
 - Supply may not be very elastic
 - Workers could bear some burden

Tax Incidence

- Corporate **capital owners** may also bear some burden
 - In the short run, capital may not be very mobile
 - Capital would bear a burden initially
 - In the longer run, it could move elsewhere
 - Increases elasticity
 - Burden on capital owners would be lower
- If capital is more mobile than labour in the longer run, then **workers would bear more of the burden**

Tax Incidence

- The corporate tax can also affect investment decisions
 - The tax lowers the after-tax return
 - Investors would move to the non-corporate sector seeking higher returns
 - Increases supply of capital in non-corporate sector
 - Leads to lower returns there too
- Investors in the **non-corporate sector** would therefore also bear some burden

Excess Burden

Distortions from Corporate Taxes

Corporate taxes create a distortion in investment decisions:

- Investors may shift away from corporate investments and toward other investments
 - Firms may choose to not incorporate despite the benefits
 - Creates a **substitution effect**
 - Shifts the allocation of capital in the economy away from the efficient level
-
- The distortion leads to an **excess burden**
 - The loss in economic efficiency beyond the tax revenue raised
 - Estimates suggest that for each dollar of corporate tax revenue raised, there is an additional **50 cents** of excess burden

Think About It

If the corporate tax creates roughly 50 cents of excess burden for every dollar of revenue raised, is it still worth having? What would happen to the distribution of the tax burden if we abolished it and tried to raise the same revenue through the personal income tax instead?

Effects on Behaviour

Introduction

- As we have learned throughout the course, taxes alter behaviour
- For corporate taxes we might expect three changes
 - The amount of investment in physical assets (equipment)
 - The types of investment in physical assets
 - How these investments are financed
- We will cover each in turn

Total Physical Investment

- Consider a firm investing in a new machine
- The cost of the machine in a particular year is
 - Depreciation: the direct cost of owning the machine that year
 - Opportunity cost: the return that could be earned if the money were invested elsewhere
- Can express both as a percentage of each dollar invested
 - Depreciation might be 2%
 - Opportunity cost might be 10%
- In example, only invest in new machine if $\text{return} > 12\%$

Total Physical Investment

! User Cost of Capital

The cost of ownership of an asset is called the **user cost of capital** – the sum of the direct and opportunity costs. It equals the **minimum earnings** required for each one dollar investment.

- In general you can express the user cost of capital as

$$C = r + d$$

- Where
 - C is the earnings from the investment
 - r is the opportunity cost of capital (interest rate)
 - d is the depreciation rate of the asset

Total Physical Investment

- Corporate taxes add to these costs
- Suppose the tax rate is t
- Now the *after-tax* earnings must be as large as the costs
- After-tax earnings are $(1 - t)C$, which means

$$C(1 - t) = r + d$$

Total Physical Investment

- Rearranging gives

$$C = \frac{r + d}{1 - t}$$

- The pre-tax user cost of capital is divided by (1 - t)
- This grosses up the required earnings to cover the tax paid

Total Physical Investment

- Example

- Suppose there are no taxes and $r = 10\%$ and $d = 2\%$
- The earnings required per dollar invested is $C = 0.10 + 0.02 = 0.12$ or 12%
- Now suppose there is a corporate tax of 45% ($t = 0.45$)
- The new required earnings is $C = \frac{0.10+0.02}{1-0.45} = \frac{0.12}{0.55} = 0.2182$ or 21.82%

- Need to earn more to cover the tax paid

Total Physical Investment

- Things get slightly more complicated when you consider deductions
- Suppose the firm can deduct the whole expense of the asset
- The after-tax user cost of capital is then

$$C = \frac{(r + d) - (r + d)t}{1 - t} = \frac{(r + d)(1 - t)}{1 - t} = r + d$$

- In this case, the corporate tax has **no effect** on the user cost of capital
 - Since no tax is paid on the investment

Total Physical Investment

- In reality, firms cannot deduct the full cost of an asset in the year of purchase
 - As we saw earlier, depreciation deductions spread the cost over time
- In this case, the deductibility mitigates the cost, but not entirely
- Suppose the deduction is equal to ψ of the cost

$$C = \frac{(r + d) - (r + d)\psi}{1 - t} = \frac{(r + d)(1 - \psi)}{1 - t}$$

Total Physical Investment

- There may also be investment tax credits (ITCs) that further mitigate the cost
- Suppose the ITC is equal to k of the cost
- Then the user cost of capital is

$$C = \frac{(r + d) - (r + d)(\psi + k)}{1 - t} = \frac{(r + d)(1 - \psi - k)}{1 - t}$$

Total Physical Investment

- Finally, a firm might finance the investment with debt
- Interest payments are deductible from taxable income
- Suppose the interest rate is λ
- The user cost of capital is then

$$C = \frac{(r + d) - (r + d)(\psi + k + \lambda)}{1 - t} = \frac{(r + d)(1 - \psi - k - \lambda)}{1 - t}$$

Total Physical Investment

Key Takeaways on User Cost of Capital

- Corporate taxes generally **increase** the user cost of capital
 - This **reduces** the incentive to invest in physical assets
 - Deductions, credits, and interest deductions can **mitigate** this effect
 - The overall impact depends on the specifics of the tax system and firm behaviour
-
- Evidence supports the idea that tax incentives can affect investment

Types of Investment

- Tax incentives are not necessarily applied equally to all investments
 - In Canada manufacturers can use [full expensing](#)
 - Also full expensing for clean energy investments
 - Other eligible property allowed accelerated depreciation
- This creates incentives to invest in certain types of assets over others
- Table on next slide shows [Marginal Effective Tax Rates](#) for various types of investments in Canada

Types of Investment

TABLE 21.4

Marginal Effective Corporation Tax Rates in Canada, 1997 and 2020: (a) by Industry and (b) by Asset Type

Industry	1997	2020
Forestry	31.6%	7.2%
Manufacturing	37.5	7.4
Construction	48.8	20.8
Transportation	41.3	14.8
Communication	54.2	15.9
Wholesale trade	47.4	21.1
Retail trade	48.0	22.9
Other services	50.0	20.4
Asset Type		
Buildings	47.8	20.8
Machinery	43.6	8.4
Land	23.9	12.4
Inventory	42.4	24.2
TOTAL	44.3	15.6

Sources: Chen and Mintz. (2006); Jack Mintz and Philip Bazel, "2020 Tax Competitiveness Report: Canada's Investment Challenge," *SPP Research Paper* 14(21) (2021), https://www.policyschool.ca/wp-content/uploads/2021/09/FMK2_2020-Tax-Competitiveness_Bazel_Mintz.pdf.

Financing Investments

- Firms need to decide to pay dividends or retain earnings
- Also how to finance an investment
 - Debt (loans, bonds)
 - Equity (issuing stock)
- The tax system can affect these

Financing Investments

- We covered earlier that preferential tax treatment of capital gains creates an incentive to retain earnings
 - If \$1 of earnings increases share value by \$1
 - Gains on shares only taxed when sold
 - And only taxed on half the gains
- So the tax system creates an incentive to **retain earnings** rather than pay dividends
- Firms do retain substantial earnings, but also pay some dividends
- Why would they do that when there is a tax incentive to retain?

Financing Investments

- One reason is **signalling**
 - Paying dividends can signal to investors that the firm is doing well
 - Investors may view dividend-paying firms as more stable and reliable
 - This can increase demand for the firm's stock, raising its price
- Second reason is differences in marginal tax rates
 - Some shareholders may have lower marginal tax rates
 - So dividends could offer a lower tax burden for them
- Investors tend to flock to firms that offer the best financial policies
 - Called **cliente effects**

Financing Investments

- The tax system can also affect how firms finance investments
- Interest payments on debt are **deductible** from taxable income
- Dividends paid to shareholders are **not deductible**
- This creates an incentive to finance investments with **debt rather than equity**
 - Raising a dollar through debt is cheaper than raising a dollar through equity
- There are additional costs to debt financing (e.g., bankruptcy risk) that increase the non-monetary cost
 - So we would not expect to see everyone using only debt financing

Discussion

The tax system encourages firms to finance with debt over equity, since interest is deductible but dividends are not. During the 2008 financial crisis, high levels of corporate debt amplified the downturn. Should the government remove the tax preference for debt to reduce financial fragility, even if it raises the cost of investment?

Taxation of Multinationals

Introduction

- Canadian firms operate in most foreign countries
- How does the corporate tax system handle income earned abroad?
- We cover that in this section
 - Note that we do not cover the taxation of foreign firms operating in Canada
- First cover the rules and then tax policy

Rules for Taxation of Foreign Income

- General principle is to be **neutral** with respect to where income is earned
 - Wants firms to face similar taxes here and abroad
- But also wants to ensure Canadian government collects tax revenue
 - Avoid **tax avoidance** which erodes the tax base
- Rules differentiate between active and passive income
 - Active income: business income from operations
 - Passive income: investment income (dividends, interest, royalties)

Rules for Taxation of Foreign Income

Passive income

- Included in taxable income when it is earned
- Taxed at the Canadian corporate tax rate
- Foreign tax credits available for taxes paid abroad

Active income

- Of incorporated subsidiaries: taxed only in the source country
 - These are separate legal entities from the main business
- Of branches: taxed in Canada when earned
 - Branches are not separate legal entities

Rules for Taxation of Foreign Income

- Corporate taxation of a firm operating in multiple countries can be problematic
- Generally, Canadian firms use the **arm's length principle**
 - Profits are each entity's own sales minus costs
- Can lead to **transfer pricing** issues
 - Firms may manipulate prices of goods/services traded between entities to shift profits to low-tax jurisdictions
 - Example: IP owned in low-tax country, charged high royalties to subsidiaries in high-tax countries
 - Royalties are tax deductible
 - Income is earned in low-tax country

Tax Policy

- We discuss two policy goals of the Canadian government
 - Maximize world income
 - Maximize national income
- First imagine the goal is to maximize world income
- Firms maximize global incomes when pre-tax returns are the same across countries
 - Want capital to move where it is most productive
 - This only happens when pre-tax returns are equalized
 - There are no distortions from taxes

Tax Policy

- Equal pre-tax returns happen when

$$r_f = r_c$$

- When taxes exist, firms distort their decisions and equalize after-tax returns

$$(1 - t_f)r_f = (1 - t_c)r_c$$

- If goal is equal pre-tax returns, then need to set

$$t_c = t_f$$

Tax Policy

- Means Canadian and foreign taxation needs to be equal
 - So no distortions are created
- Canadian firms allow a tax credit up to the Canadian tax on foreign income
 - Cannot allow more than that or foreign firms could increase taxes to raise revenue without any consequence
- The government may instead try to **maximize national income**
 - Focus on income that accrues to Canadian residents
 - Equals before tax income earned in Canada plus after-tax income earned abroad

Tax Policy

- In this instance, policy should be to equalize

$$r_f(1 - t_f) = r_c$$

- Means pre-tax Canadian returns equal after-tax foreign returns
 - Prioritizes income earned in Canada
 - Foreign income pre-tax needs to have higher return than Canadian

Tax Policy

- If Canada wants to maximize national income, then should set

$$r_f(1 - t_f)(1 - t_c) = r_c(1 - t_c)$$

- Sets the same tax rate on after-tax foreign income as on pre-tax Canadian income
 - Can do this by allowing **foreign tax deduction**

Corporate Tax Reform

Introduction

- The current corporate tax system in Canada is **complex**
- The treatment of dividends is in particular complicated
 - Taxed at both corporate and personal levels
 - Complex system of gross-ups and tax credits to mitigate double taxation
- There have been calls for reforming the corporate tax system
 - Simplify the system
 - Improve efficiency and equity
- We discuss a few of the current ideas

Partnership Method

Partnership Method

The **partnership method** taxes income only once at the personal level – removes corporate taxation altogether and taxes shareholders on their share of corporate income at personal income rates.

- Some benefits include
 - Less excess burden by removing corporate tax distortions
 - Eliminate misallocation of capital between corporate and non-corporate
 - Lower tax on capital overall
 - Less incentive for retained earnings
 - Less incentive for debt financing
 - More saving due to lower capital taxes
 - Increases investment and economic growth

Partnership Method

- Some potential issues
 - Administratively complex
 - Tracking income for many shareholders
 - How to treat short term shareholders
 - Problems with distribution of income
 - People with high personal tax rates benefit less
 - Benefit people with more income from capital more

Other Methods

Dividend Exclusion

- Make dividends non-taxable at personal level
- Only taxed at corporate level

Dividend Deduction

- Allow corporations to deduct dividends paid from taxable income
- Only taxed at personal level

Dual Income Tax

- Capital income taxed at lower rate to encourage investment
- Labour income taxed at higher rate for progressivity

Summary

Summary

- Corporations are taxed separately from individuals
- Depreciation, interest, and dividends have special tax treatments
- Corporate taxes create incidence and excess burden issues
- Taxes affect investment decisions and financing choices

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